

ME350 FA 2025 Section 002

Lab #8

Monday 12:30 PM - 2:30 PM

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Reminders

- Viewing the Mechatronics Lab orientation video was **required by today!**
- **Mid semester peer evaluation is due Wednesday, Oct 22 at midnight**
- Homework 5 is due Thursday, October 23th
- **Lab 8 Assignment** due at the start of next lab
- DR 2 assignment has been posted and is due Nov 3 (Thursday labs) or Nov 4 (Friday labs) or Nov 5 (Monday labs)



Calendar

Oct 19	Oct 20	Oct 21	Oct 22	Oct 23	Oct 24
		Lecture 15 - Sensors Part 1 (JK)		Lecture 16 - Sensors Part 2 (JK) - Homework 5 Due (Transmissions 2) - Homework 6 Released	
	Lab 8 - Sensors and Microcontrollers 1: Wiring, switches, and IR Prox sensor Lab Assignment #7 Due - Manufactured GR2 Parts		Mid semester Peer Evaluation Due	Lab 9 - Sensors and Microcontrollers 2: Motors and Encoders Lab Assignment #8 Due - Lab 8 Worksheet with screenshots	Lab 9 - Sensors and Microcontrollers 2: Motors and Encoders Lab Assignment #8 Due - Lab 8 Worksheet with screenshots
Oct 26	Oct 27	Oct 28	Oct 29	Oct 30	Oct 31
		Lecture 17 - Signals Part 1 (JK)		Lecture 18 - Signals Part 2 (JK)	
	Lab 9 - Sensors and Microcontrollers 2: Motors and Encoders Lab Assignment #8 Due - Lab 8 Worksheet with screenshots			Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due	Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due
Nov 2	Nov 3	Nov 4	Nov 5	Nov 6	Nov 7
		Lecture 19 - Controls Part 1 (JK)		Lecture 20 - Controls Part 2 (JK) - Homework 6 Due (Sensors) - Homework 7 Released	
	Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due			Lab 11 - Sensors and Microcontrollers 4: State Machines Lab Assignment #10 Due	Lab 11 - Sensors and Microcontrollers 4: State Machines Lab Assignment #10 Due
	DR 2 DUE TODAY BY MIDNIGHT (Thursday Section)	DR 2 DUE TODAY BY MIDNIGHT (Friday Section)	DR 2 DUE TODAY BY MIDNIGHT (Monday Section)		

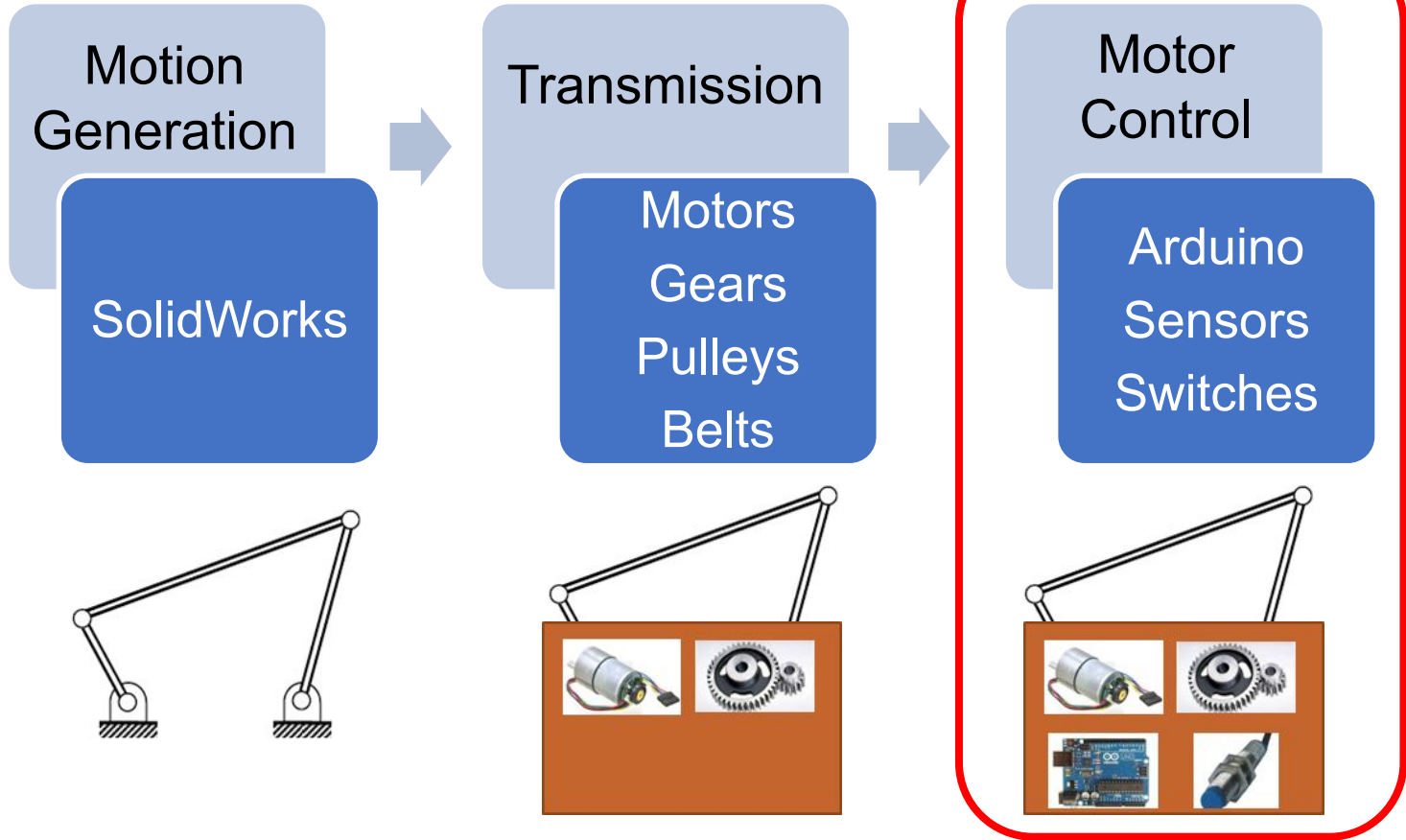


Project Schedule

- Lab 8 - Linkage + transmission + limit switch must be completely assembled!!
(You'll need it for Sensors/Controllers Lab 2)
- Lab Assignment #8 (Working circuit from Sensors/Controllers Lab 1) due by start of Lab 9
 - Keep the wiring together since each lab builds on the previous one



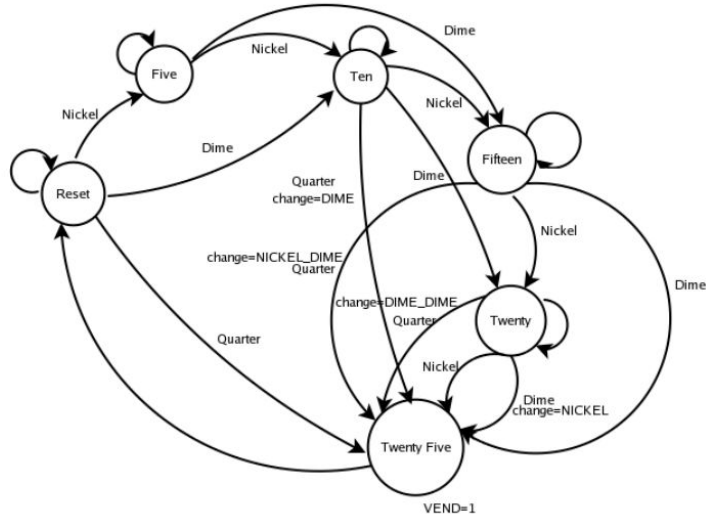
Where Are We Now?



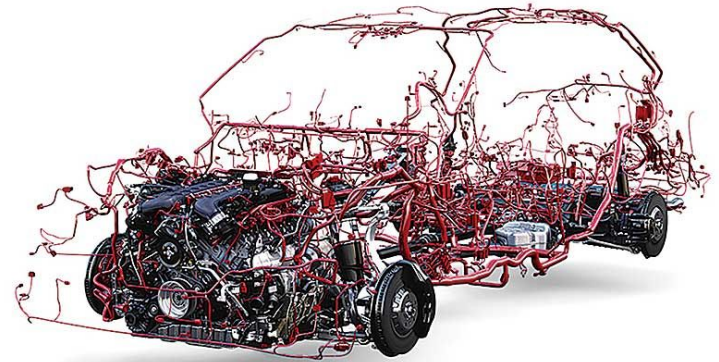
Mechatronics

Mechatronics in the ME350 project:

- Electrical: Wiring sensors and the motor
- Programming: Finite State Machines
- Control: PID feedback control

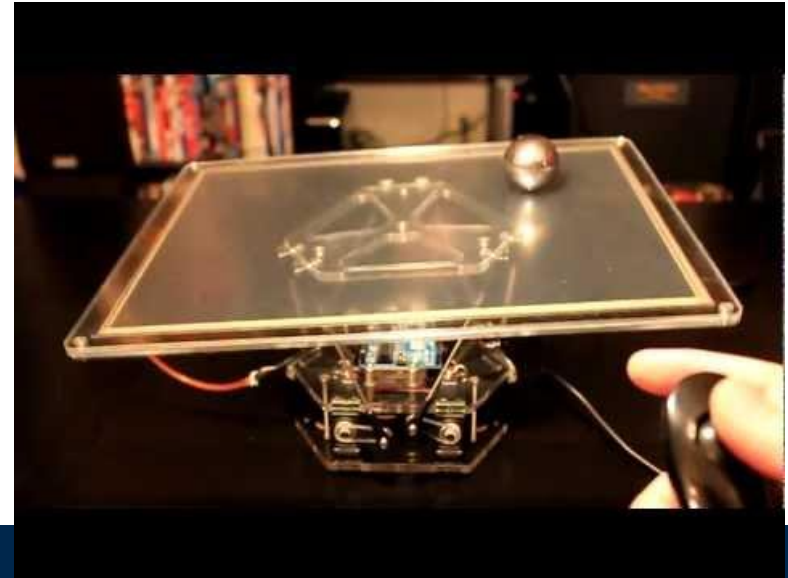


Vending Machine Finite State Machine (Ajay Sharma)
<http://www.csun.edu/~ags55111/doc/526/526report.pdf>



Bentley Bentayga Wiring Harness:

<https://www.motor1.com/news/65202/bentley-bentayga-wiring-harness-is-weirdly-beautiful/>



Mechatronics Lab Overview and Safety Training

- **Top 3 Lab Rules**

- No working alone
- No food or drink in the lab – for safety and cleanliness
- No equipment or tools leave the lab

- **Lab Etiquette**

- Leave lab cleaner than you found it
- Take ownership – put tools away, recycle any/all components (wire, resistors, etc.) unless specified
- Many users – all contribute to organization
- Contact GSI or lab manager if issues when you come to use lab

- **Location of critical items**

- First Aid kits – front of the class
- Fire extinguisher, phone, etc.
- Tool box – each station has a set of tools – team #, lab station # = tools set #, return to tool box

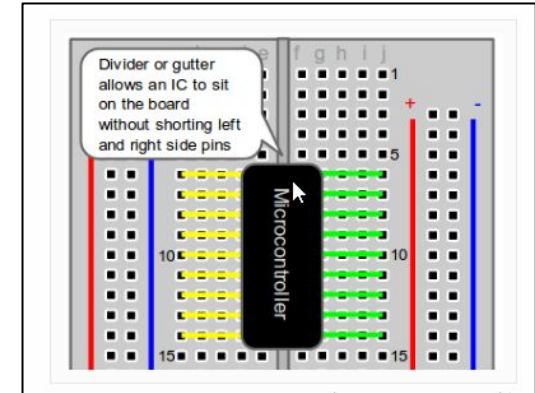
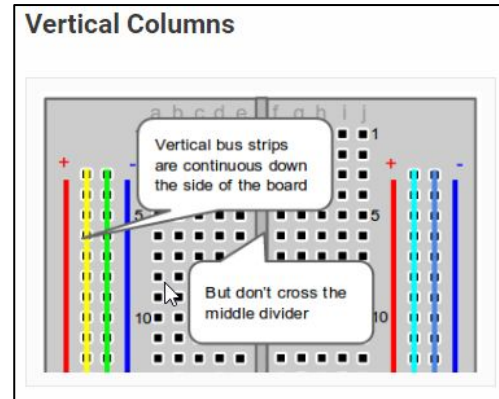
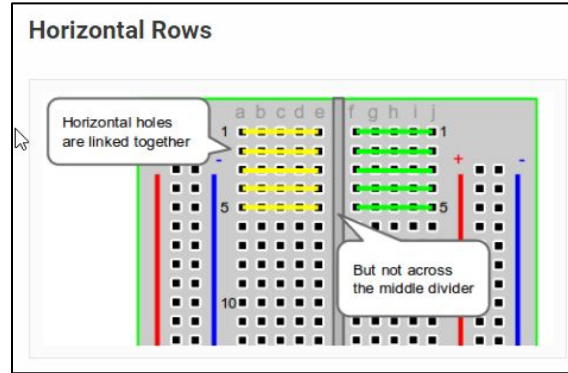
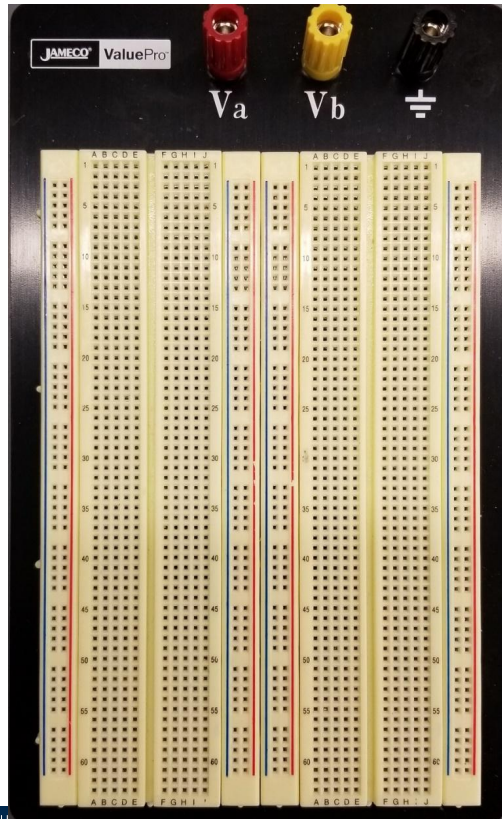
- **Workstations**

- Computers – have all software you should need, do not alter, add, change cables, etc.
- USB docks for MyRio or Arduino connection
- Use Google Drive or N: drive for files/file storage – do not store anything local to computer



Mechatronics Lab Overview and Safety Training

- Breadboard basics

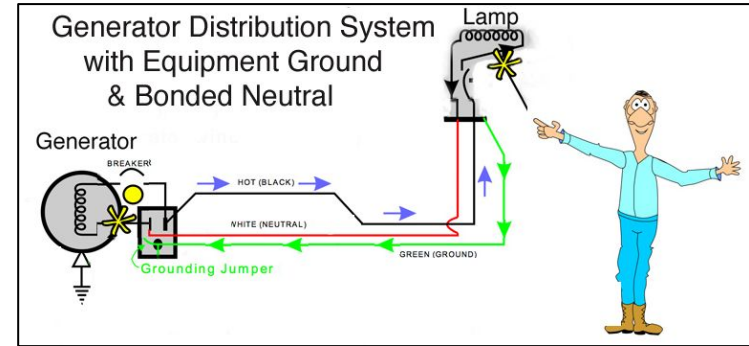
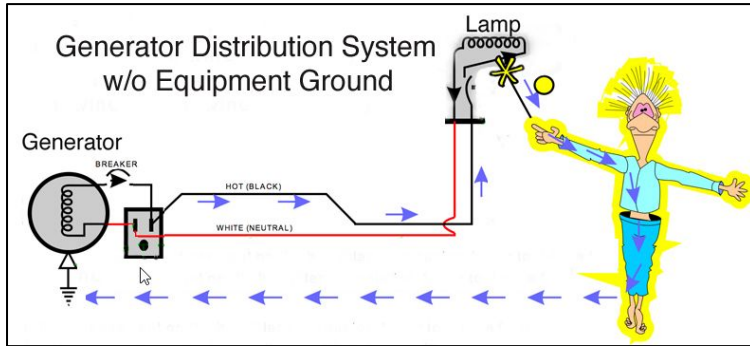


<https://computers.tutsplus.com/tutorials/how-to-use-a-breadboard-and-build-a-led-circuit--mac-54746>

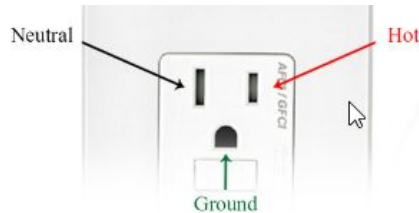
Mechatronics Lab Overview and Safety Training

• Understanding Ground

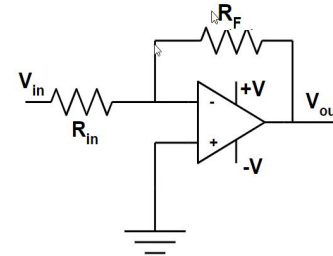
- Electricity follows path of least resistance, power supplies want to be at lowest energy state = positive loves ground!
- Follow instructions – electricity can harm you!



Always have the
Power Strip off
when building
Circuits



Understand:
Earth Ground vs Local
Reference

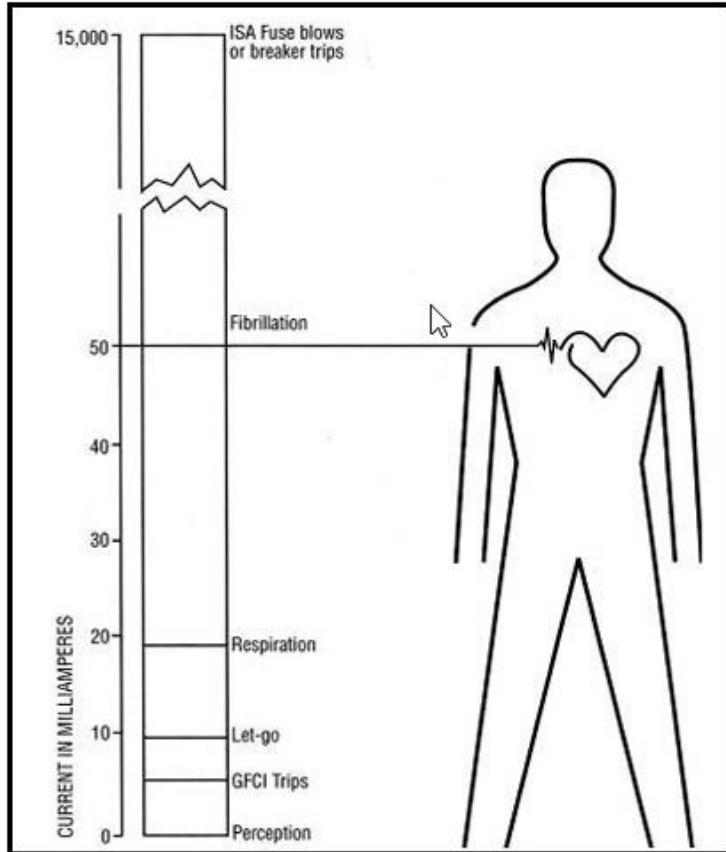


Local Ground (Not the
same as earth ground).
The ground your circuit
references in Lab.

Earth Ground (Grounding Rod/Plate for GGB)

Mechatronics Lab Overview and Safety Training

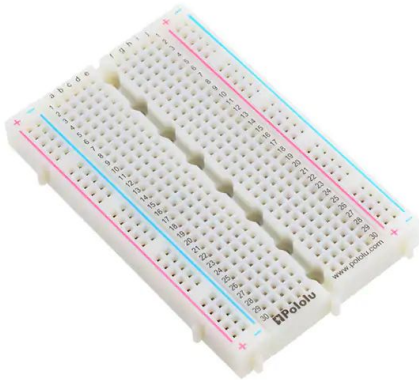
Anything above 8mA is unsafe. There are 15A = 15,000mA from the wall



Readings		Effects
Safe Current Values	1 mA or less	Causes no sensation - not felt.
	1 mA to 8 mA	Sensation of shock, not painful; Individual can let go at will since muscular control is not lost.
Unsafe current values	8 mA to 15 mA	Painful shock; individual can let go at will since muscular control is not lost.
	15 mA to 20 mA	Painful shock; control of adjacent muscles lost; victim can not let go.
	50 mA to 100 mA	Ventricular fibrillation - a heart condition that can result in death - is possible.
	100 mA to 200 mA	Ventricular fibrillation occurs.
	200 mA and over	Severe burns, severe muscular contractions - so severe that chest muscles clamp the heart and stop it for the duration of the shock. (This prevents ventricular fibrillation).

Wiring and Testing Sensors

- Wire sensors to the Arduino microcontroller
 - Proximity Sensor
 - Encoder
 - Toggle Switch
 - Limit Switch
- Test them using the provided scripts
- Document your wiring



Mechatronics Labs

- **You are encouraged to work ahead in the labs! The sooner you finish the tutorials, the more time you have to tune before final testing!**
- Part 1 (today) – Integrate limit switch, toggle switch, proximity sensor
- Part 2 – Integrate motor and encoder
- Part 3 – Implementing PID control
- Part 4 – Making system autonomous (state machine)



Examples

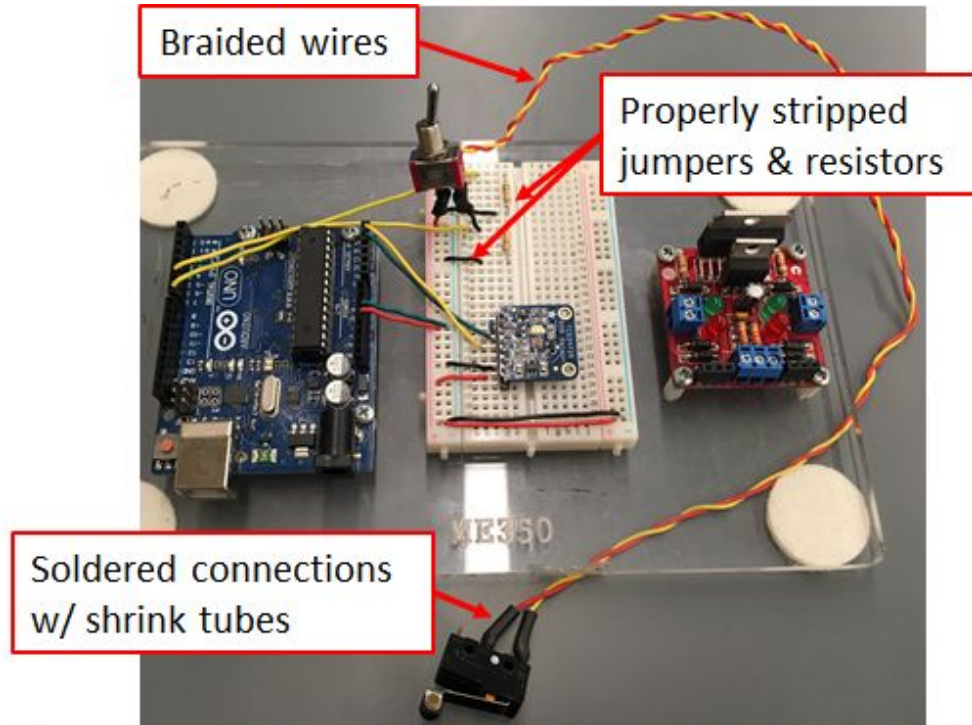
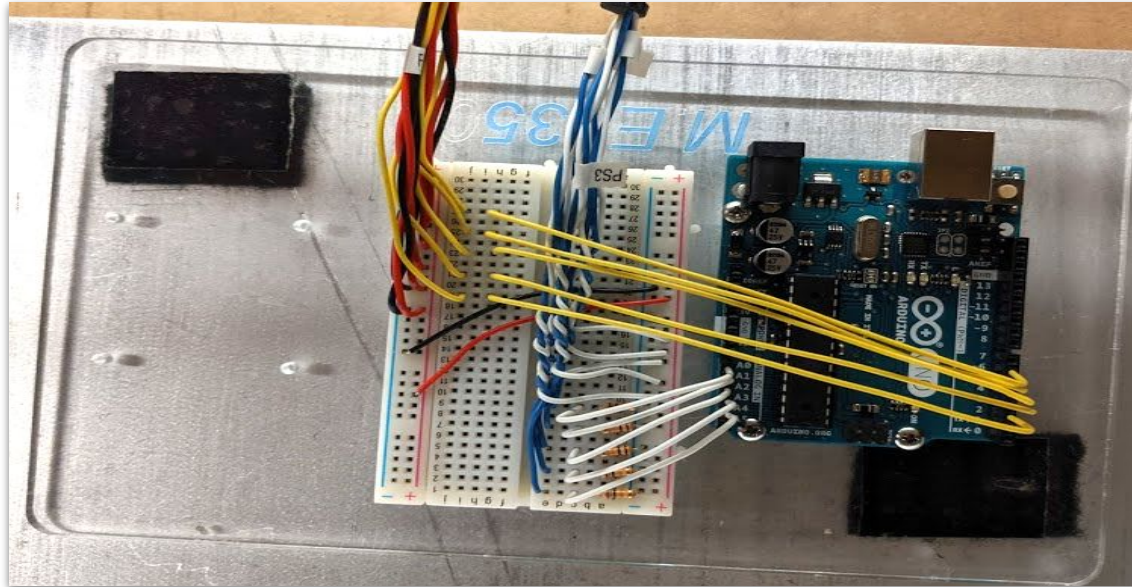


Figure 1: This is okay but not the best. Your wiring can be different!

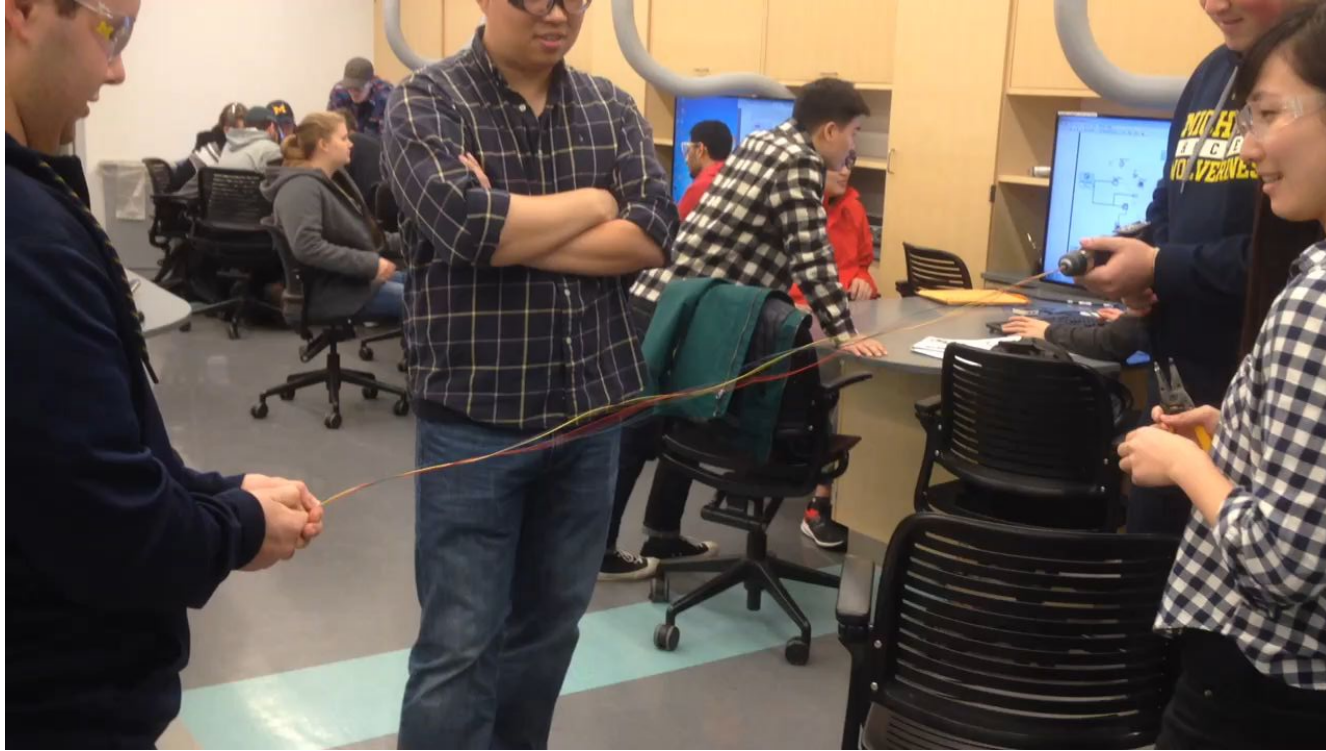
Examples

Playing Field Wiring:

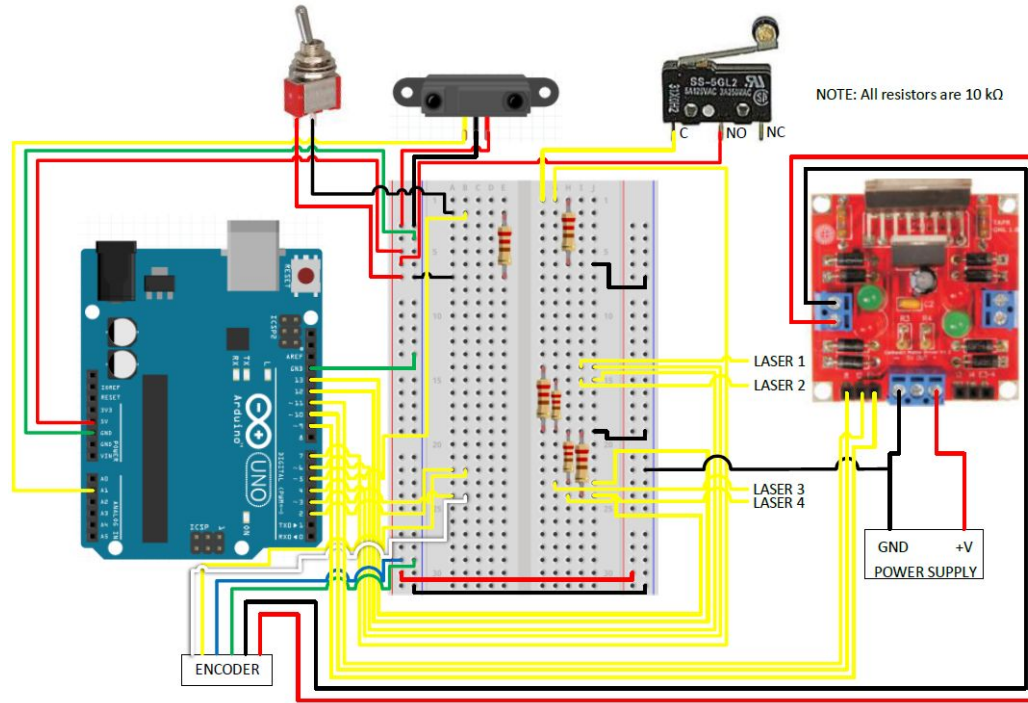
- This is a good example -- your wiring should be similarly neat



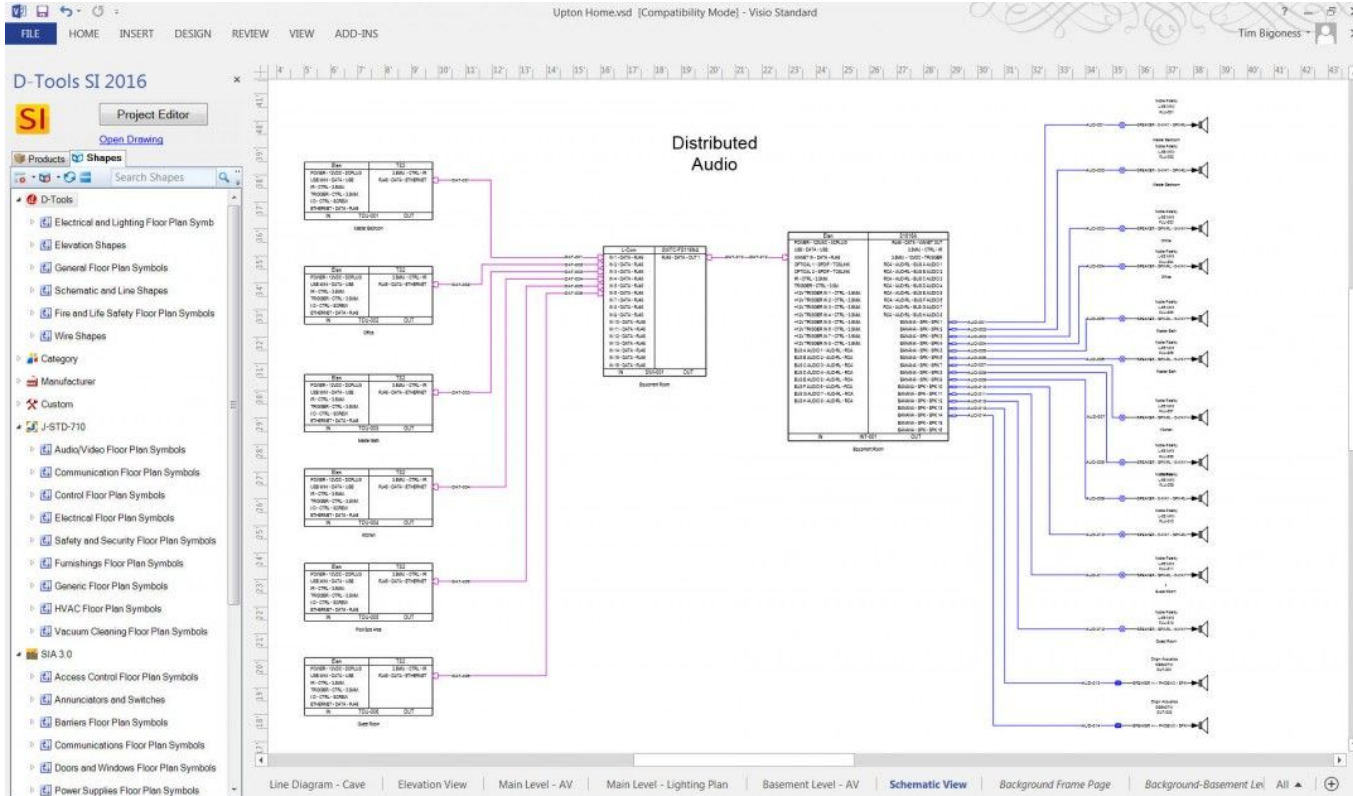
Ex: Wire “Braiding”



Ex: Fritzing - available on Mechatronics Computers for Free



Ex: Visio



Arduino Tips

To prevent Arduino from losing connection to COM port:

- Close serial monitor before downloading sketches
- Don't have two Arduino sketches open at the same time
- Don't open the serial monitor open until the sketch is downloaded
- Note that you may have to plug into a different COM (USB) port if your current COM port is lost
- Hit the reset button on the arduino to restart the program without having to re-upload

If you're using the Mechatronics Lab computers, SAVE YOUR ARDUINO CODE ON Google Drive or a flash drive!!



Today

At the minimum, work through Lab 8 Tutorial:

- This is the start of your linkage wiring!
 - Make it neat and easy to follow
 - You will keep this together and build off of it for all labs and testing to come
- The mechatronic lab is self-paced. The sooner you get all the tutorials done, the more time you will get to test and tune
- **SIGN THE CLEANUP CHECKLIST**

What you should do before next lab:

- Complete mechanism (including transmission!)
- Complete Lab 8 assignment



Today

- Run through [Lab 8 Tutorial](#):
 - Run test scripts
 - Fill out lab 8 assignment

Due Next Lab (ALL IN LAB 8 ASSIGNMENT)

1. Implement a custom print message for your test sketches
2. Submit screenshots of the serial monitor with your custom messages in the [worksheet](#)
3. Wiring Diagram from Lab 8 due
4. Linkage + transmission + limit switch must be completely assembled!! (You'll need it for Lab 8)

